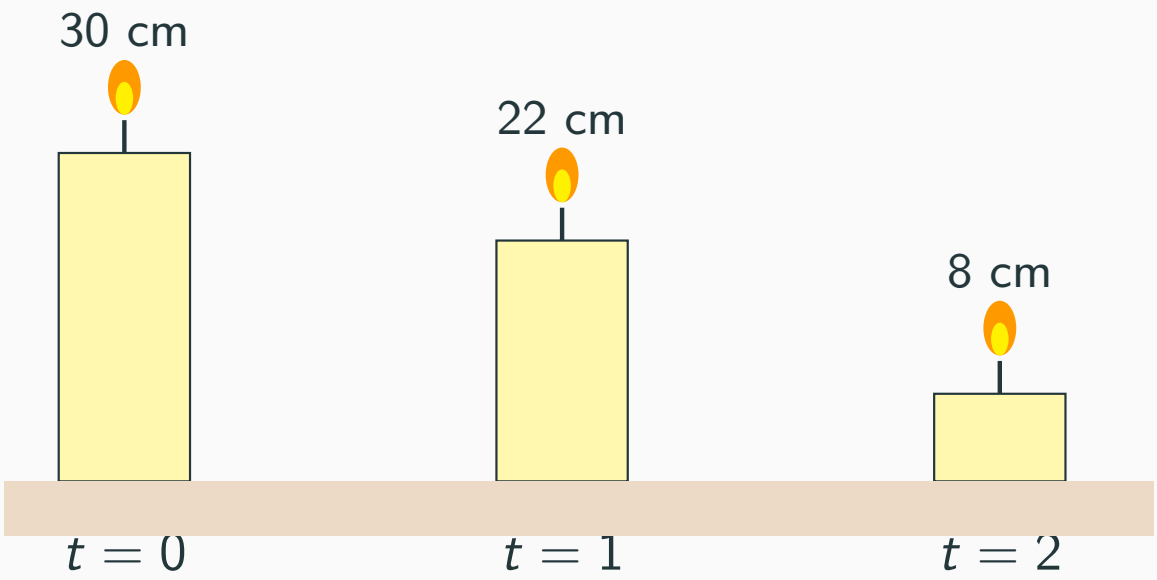
Lecture 11: Linear Regression

MATH203 Applied Linear Algebra

The Candle Problem

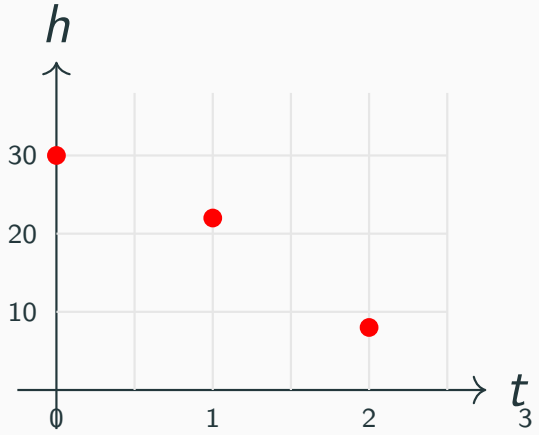
A Burning Candle

You light a candle and measure its height over time.



The Data

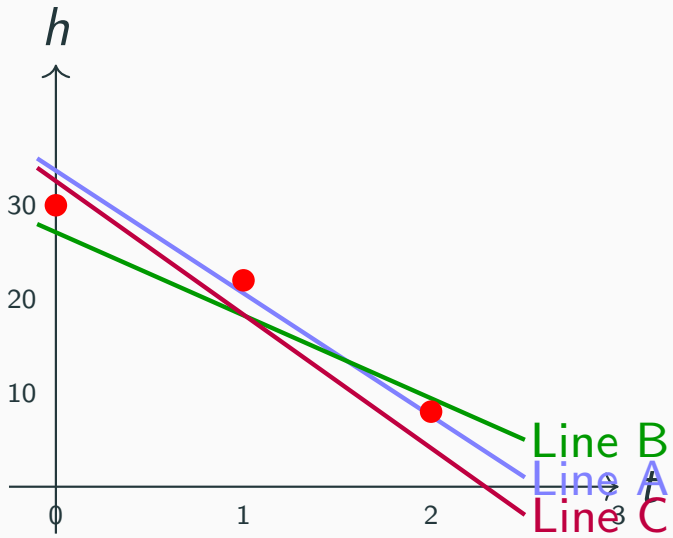
Time (hours)	Height (cm)
0	30
1	22
2	8



The candle gets shorter over time. **Can we predict its height at $t = 3$?**

Drawing a Line

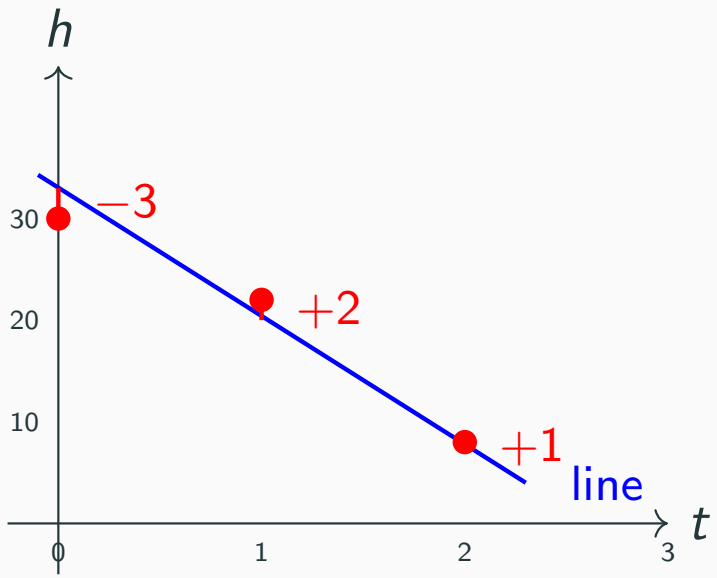
Our instinct: draw a straight line through the data.



Three different lines. Which one is the **best**?

What Does "Best" Mean?

Pick a line (not a great fit). Measure the **vertical gap** at each data point.



Three gaps: -3, $+2$, $+1$.

The Error

How bad is this line? Add up the squared gaps:

$$\text{Error} = (-3)^2 + (+2)^2 + (+1)^2 = 9 + 4 + 1 = 14$$

A different line would give different gaps, and a different error.

Goal: Find the line whose error is the **smallest possible**.

How?

We could try thousands of lines and pick the one with the smallest error.

But there is a **geometric shortcut**.

We will translate this problem into a picture in $\mathbb{R}^3$, where the answer becomes **obvious**.

Data as Geometry

Three Measurements

Your experiment produced **three numbers**:

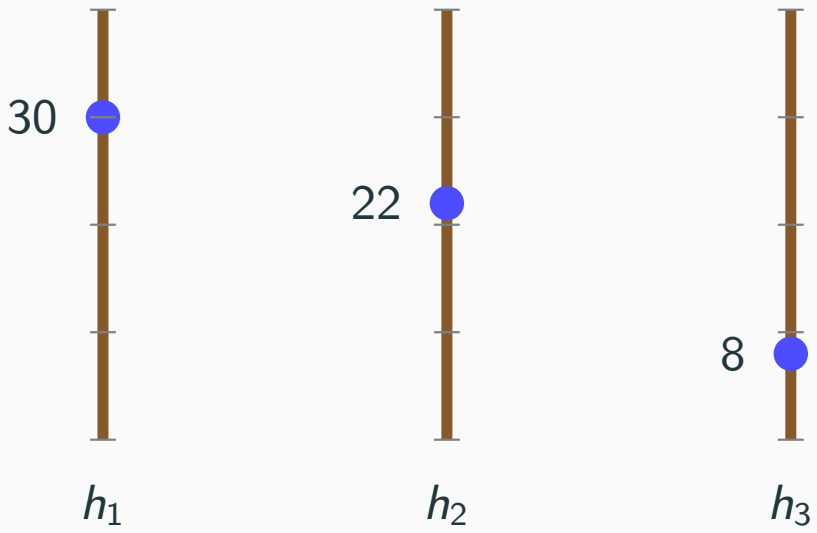
$$h_1 = 30 \quad h_2 = 22 \quad h_3 = 8$$

That is all the information you have about this candle.

Three numbers. Can we think of them as **one object**?

Marbles on Chopsticks

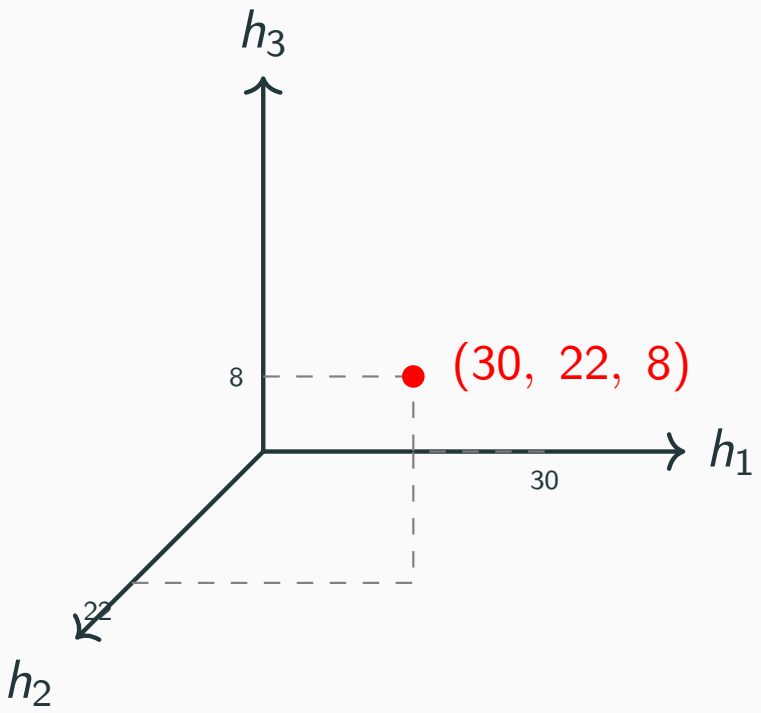
Imagine three **chopsticks** standing upright. Place a marble on each one at the measured height.



Three chopsticks. Three marbles. The **positions of all three marbles together** describe your experiment.

Three Chopsticks = Three Axes

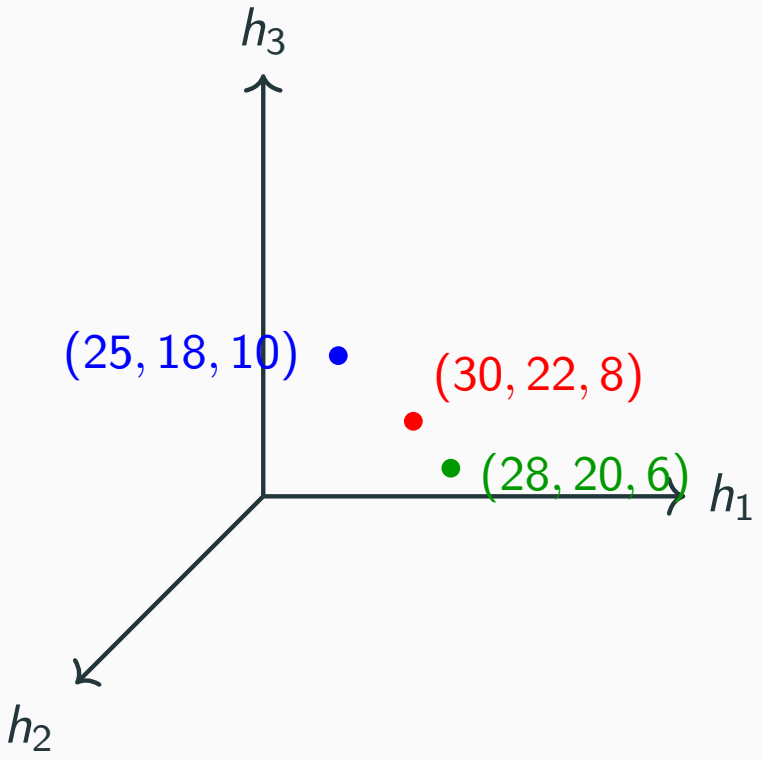
Now tilt the chopsticks into three perpendicular directions.



The three marble positions become **one point** in 3-dimensional space:
$\mathbf{b} = (30,\ 22,\ 8)^T \in \mathbb{R}^3$

Different Candle, Different Point

If you repeated the experiment with a different candle:



Every possible experiment produces some point in $\mathbb{R}^3$.

All of the candle's data lives in one place: the space $\mathbb{R}^3$.

Data = A Point

3 measurements on a number line

$$\Downarrow$$

1 point in $\mathbb{R}^3$

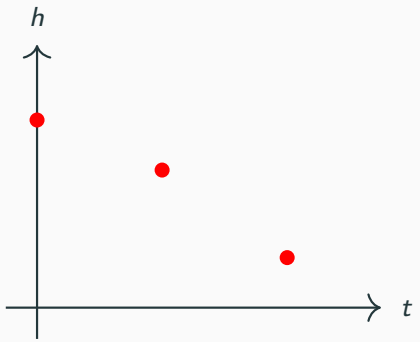
Our candle experiment gave us:

$$\mathbf{b} = \begin{pmatrix} 30 \\ 22 \\ 8 \end{pmatrix}$$

This is our **observed data**, living as a single point in $\mathbb{R}^3$.

Two Different Pictures

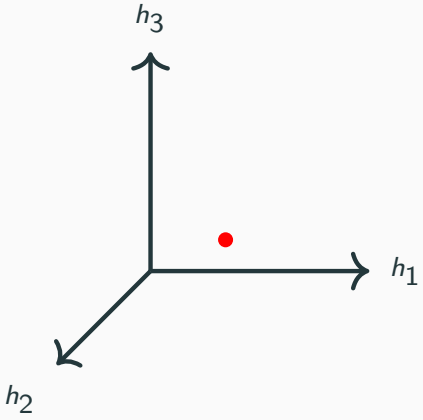
Scatter plot



3 dots in 2D

Each dot = one measurement

Data space



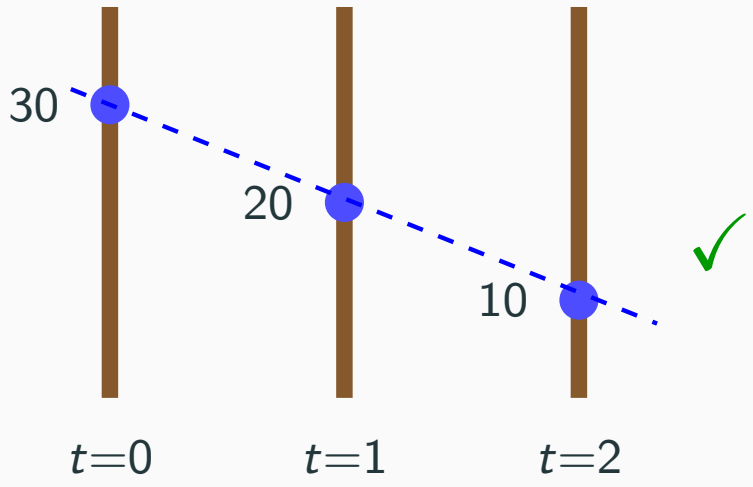
1 dot in 3D

The dot = all three measurements together

These are **two completely different spaces**. Do not confuse them.

What Is "Perfect Data"?

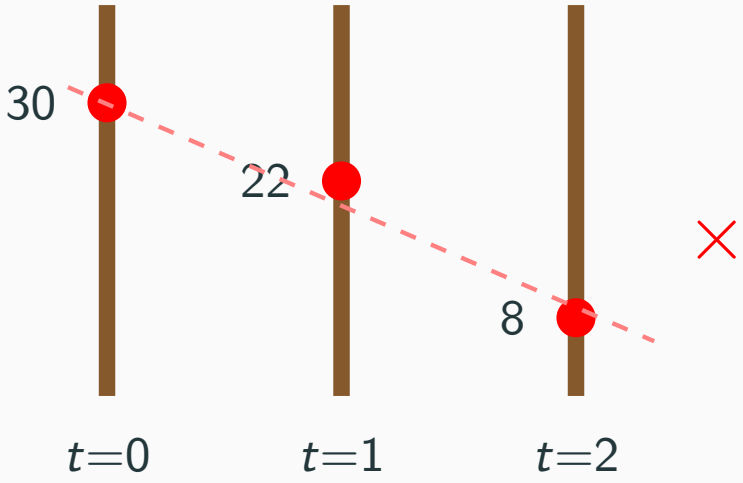
Place marbles on the chopsticks. If the marble tops form a **straight line**: perfect!



The marbles at 30, 20, 10 lie on a straight line. **Perfect data.**

Our Data Is Not Perfect

Now put our **actual** marbles on the same chopsticks:

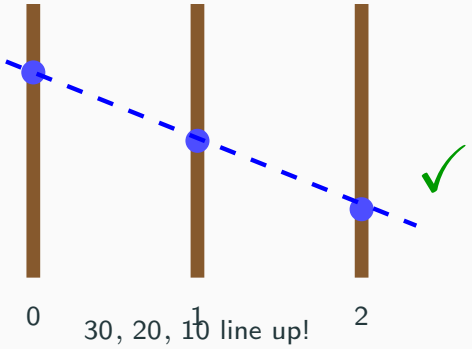


30, 22, 8 do **not** lie on any straight line. Our data is imperfect.

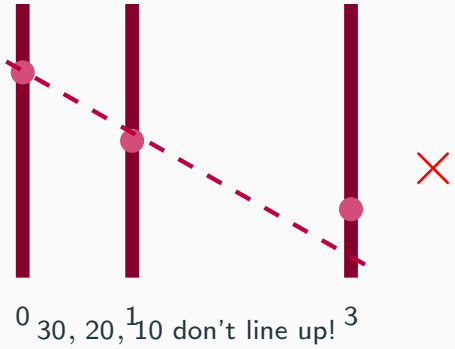
Chopstick Positions Define "Perfect"

The **same marble heights** can be perfect or imperfect — depending on **where the chopsticks are**!

Chopsticks at $t = 0, 1, 2$



Chopsticks at $t = 0, 1, 3$

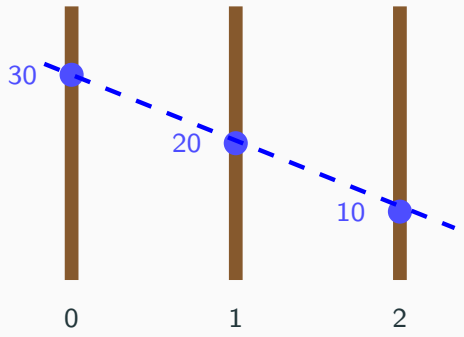


You need the chopstick positions first, then you can say what is "perfect."

Different Chopsticks, Different Perfect Data

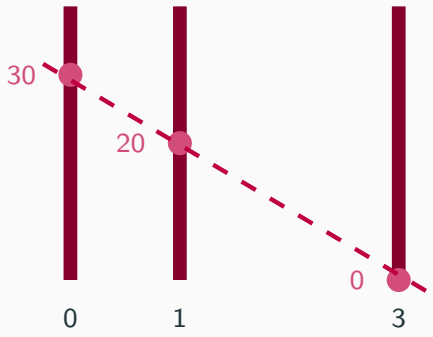
For the same burning rate, the **perfect marble heights** change when you move the chopsticks.

Chopsticks at $t = 0, 1, 2$



Perfect: $30, 20, 10$

Chopsticks at $t = 0, 1, 3$

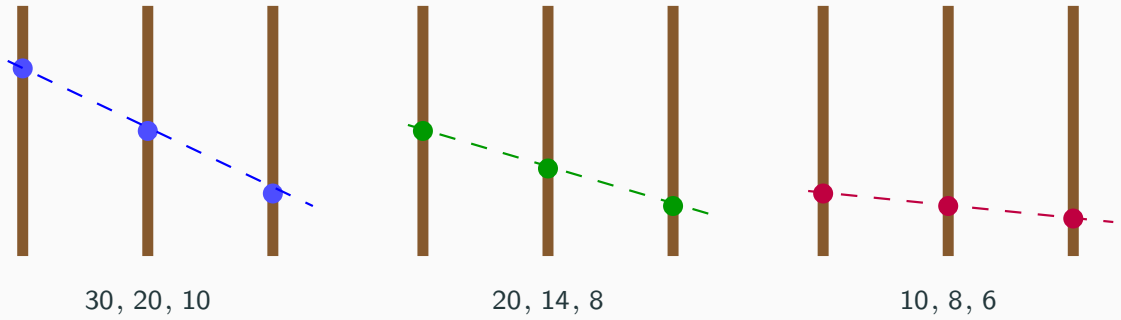


Perfect: $30, 20, 0$

Same candle, same burning rate. But the chopstick positions changed, so the perfect marble heights are **completely different**.

Many Perfect Configurations

Back to our chopsticks at $t = 0, 1, 2$. Here are several perfect marble sets:

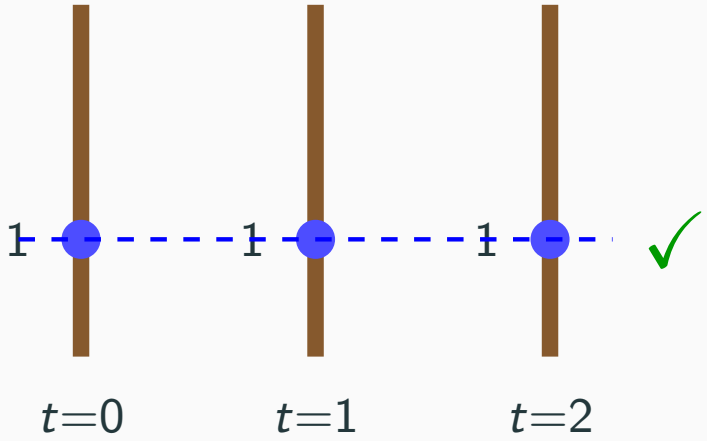


Each set of perfect marbles = one point in $\mathbb{R}^3$.

What shape do all these points form?

Special Perfect Data #1: Level Marbles

Put all marbles at the **same height** (say, height 1).

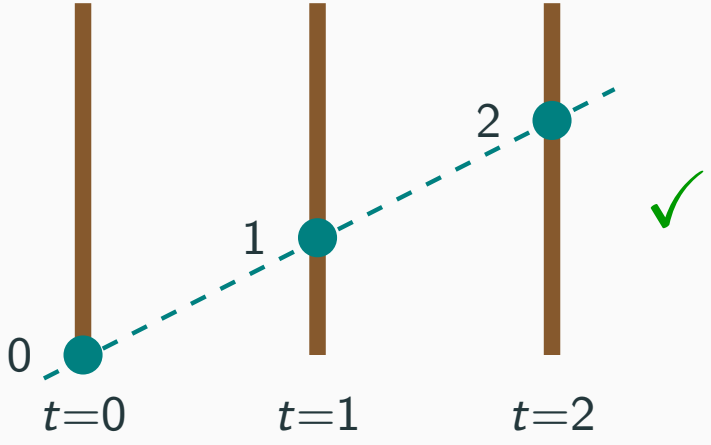


Level marbles **always** form a straight line, no matter where the chopsticks are!

In data space: the point $(1, 1, 1)$.

Special Perfect Data #2: Diagonal Marbles

Put each marble at the **chopstick's own position**: height $= t$.



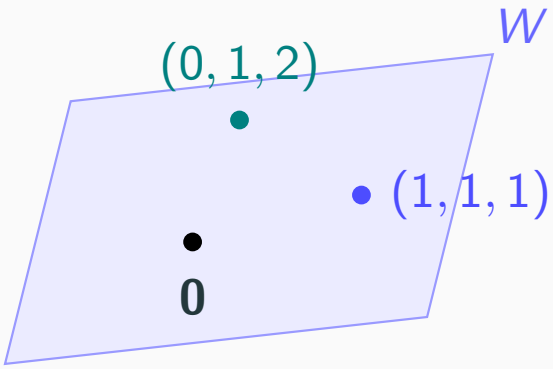
The marbles trace out the line $h = t$. Perfect!

In data space: the point $(0, 1, 2)$ — the chopstick positions themselves.

Three Points Determine the Plane

We found three perfect-data points:

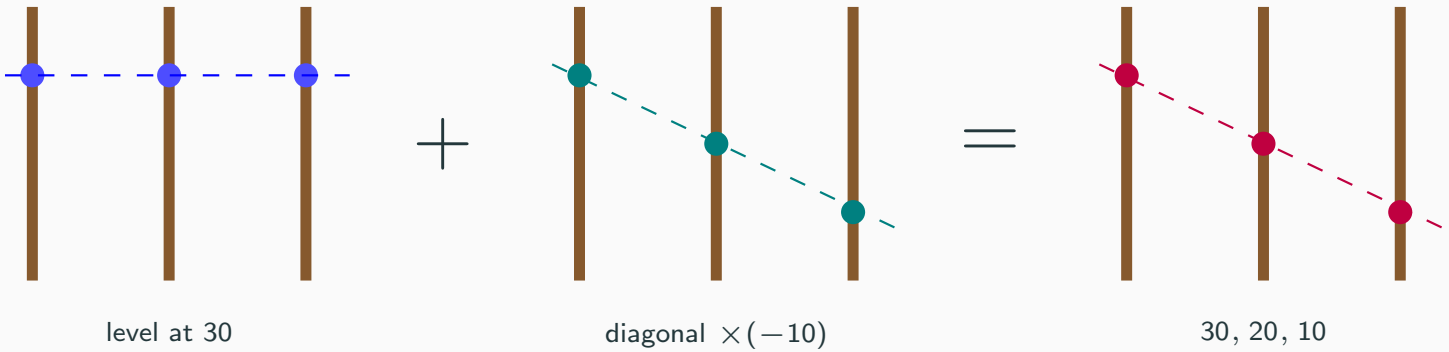
- $\mathbf{0} = (0, 0, 0)$
  no candle $\rightarrow$ trivially perfect
- $(1, 1, 1)$
  level marbles $\rightarrow$ horizontal line
- $(0, 1, 2)$
  diagonal marbles $\rightarrow$ slope-1 line



Three points that don't lie on a line determine a **plane**.

This plane W contains **all** perfect data.

Every Perfect Line = A Mix of These Two

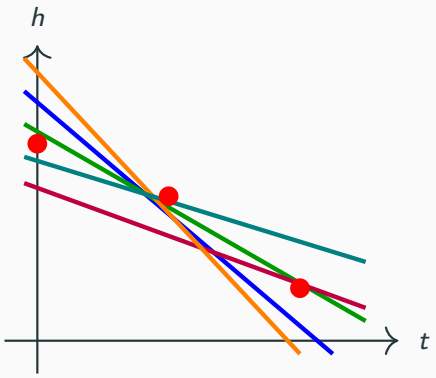


Any perfect line is a **mix** of "level" and "diagonal."

All mixes of two directions $\rightarrow$ a **plane** in data space.

Lines in Scatter Plot = Points on a Plane

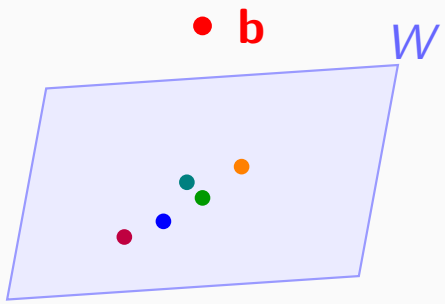
Scatter plot



Many lines, each a different "perfect candle"

$\longleftrightarrow$

Data space $\mathbb{R}^3$



Each line becomes one point on the plane W

Each line in the scatter plot is one point on the plane. Our data **b** is **off the plane**.

Our Plane

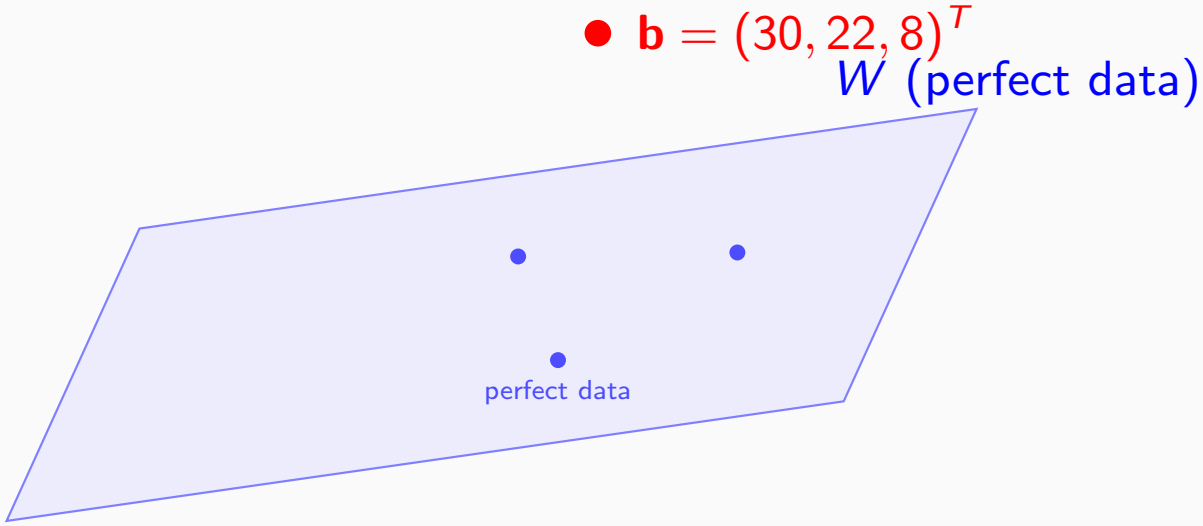
$$W = \mathrm{Col}\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$$

Column 1: $(1, 1, 1)^T$ $\leftarrow$ level marbles

Column 2: $(0, 1, 2)^T$ $\leftarrow$ chopstick positions

Move the chopsticks $\rightarrow$ change column 2 $\rightarrow$ change the plane.

The Picture



Our data **b** is **not on the plane** W (the candle didn't burn perfectly).

Model = A Plane

All perfect lines in the scatter plot

$$\Downarrow$$

A plane W in data space $\mathbb{R}^3$

Which plane? That depends on **where you put the chopsticks** (the measurement times).

Our chopsticks at $t = 0, 1, 2$ give: $W = \text{Col}\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$

The Translation So Far

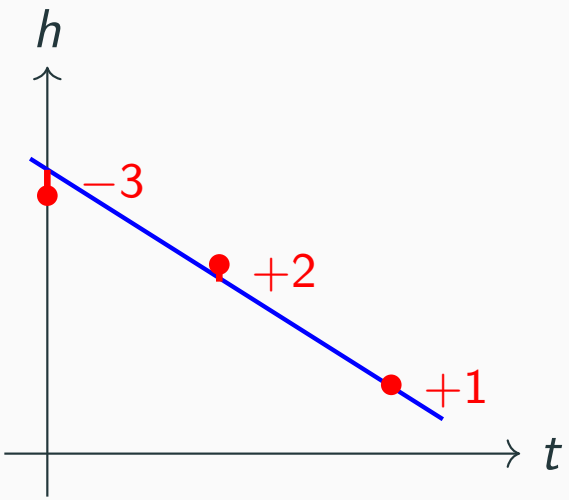
Candle Problem	**Geometry in $\mathbb{R}^3$**
Measurements $(30, 22, 8)$	A point $\mathbf{b} \in \mathbb{R}^3$
Model $h(t) = a + bt$	A plane $W \subseteq \mathbb{R}^3$
Data has errors	$\mathbf{b}$ is not on W
Find the best line	???

What does "find the best line" become in geometry?

Error = Distance

Error in the Scatter Plot

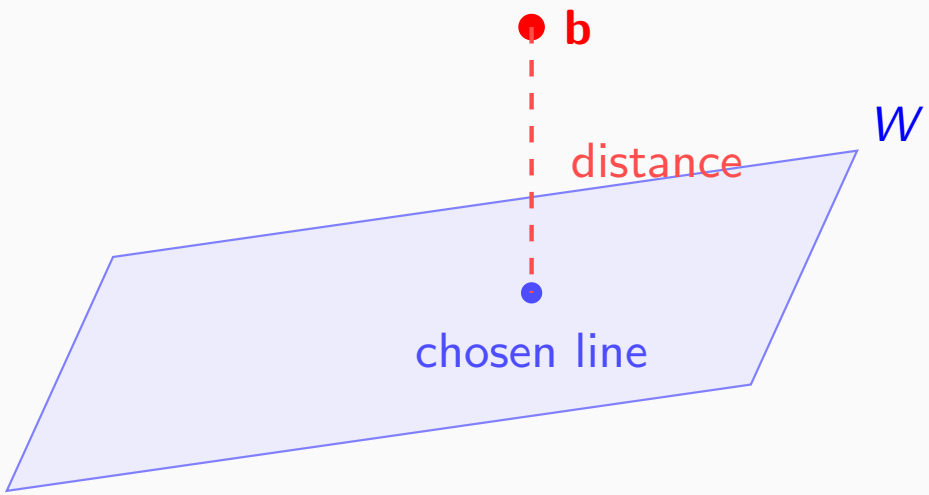
Pick a line. Measure the three gaps:



$$\text{Error} = (-3)^2 + (+2)^2 + (+1)^2 = 9 + 4 + 1 = 14$$

Distance in Data Space

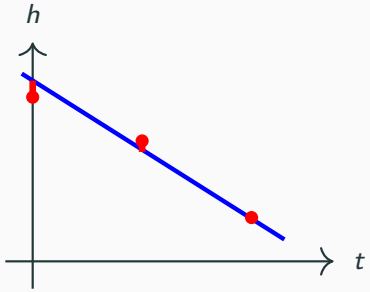
That same line is a point on the plane W. How far is it from $\mathbf{b}$?



$$\text{Distance}^2 = (30-33)^2 + (22-20)^2 + (8-7)^2 = 9 + 4 + 1 = \mathbf{14}$$

They Are the Same!

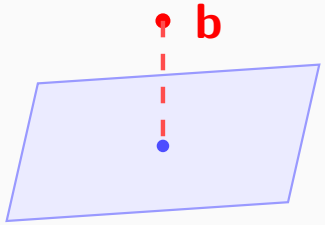
Scatter plot



$\text{gap}_1^2 + \text{gap}_2^2 + \text{gap}_3^2$
$= 9 + 4 + 1 = \mathbf{14}$

$=$

Data space



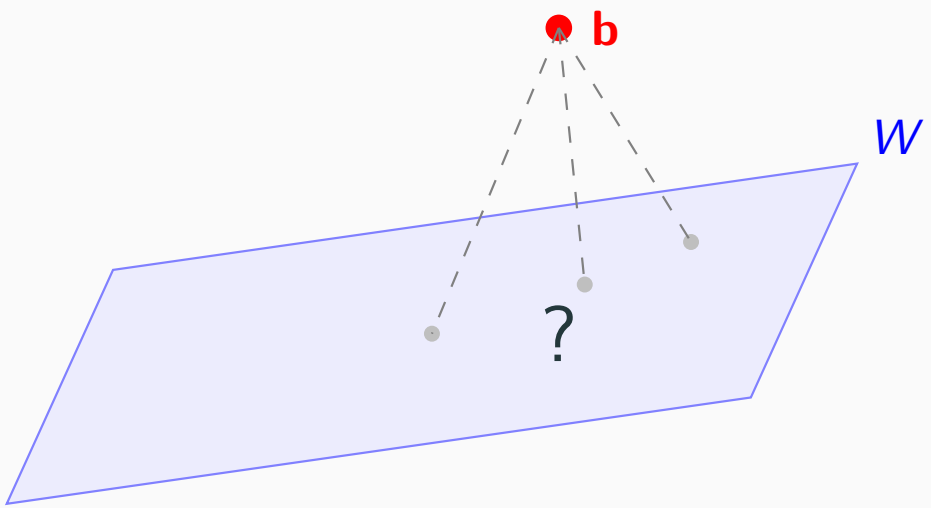
distance^2
$= 9 + 4 + 1 = \mathbf{14}$

The sum of squared gaps in the scatter plot **equals** the squared distance in data space.

(Each gap becomes one coordinate of the difference vector. Pythagorean theorem does the rest.)

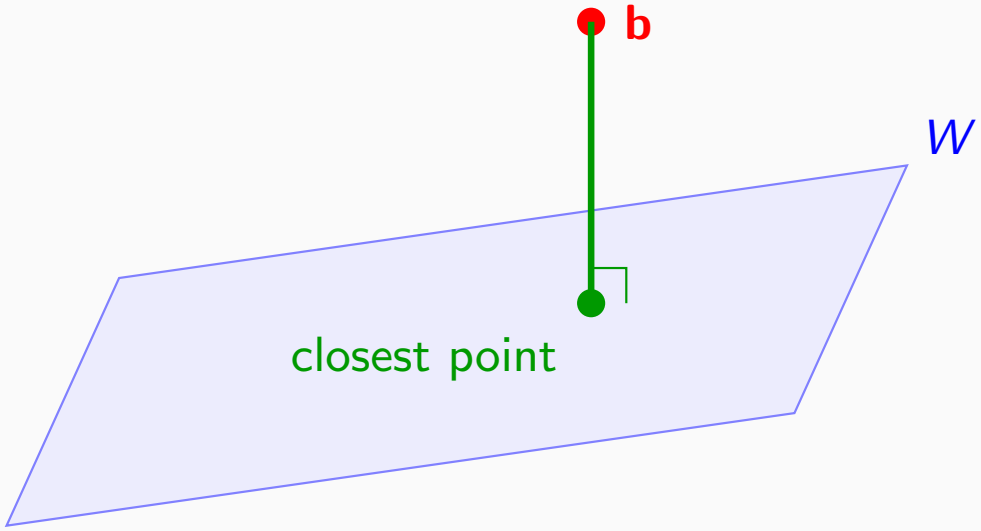
Minimize Error = Minimize Distance

Finding the **best line** means finding the point on W **closest to b**.



Which point on the plane W has the **shortest** distance to $\mathbf{b}$?

Drop a Perpendicular!



The closest point on a plane is always the **foot of the perpendicular**.

This is the **orthogonal projection** of $\mathbf{b}$ onto W.

We Already Know How to Do This!

Callback (Lecture 10): The orthogonal projection onto $\mathrm{Col}(B)$ is

$$P = B(B^T B)^{-1} B^T$$

For our candle, $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$.

The best line is encoded in $P\mathbf{b}$ — the projection of our data onto the perfect-data plane.

The candle problem is solved by a projection.

The Complete Translation

Candle Problem	**Geometry in $\mathbb{R}^3$**
Measurements $(30, 22, 8)$	A point $\mathbf{b} \in \mathbb{R}^3$
Model $h(t) = a + bt$	A plane $W \subseteq \mathbb{R}^3$
Data has errors	$\mathbf{b}$ is not on W
Sum of squared gaps	Distance^2 from $\mathbf{b}$ to W
Find the best line	**Orthogonal projection of $\mathbf{b}$ onto W**

Solving the Candle Problem

The Formula

Callback (Lecture 10): The orthogonal projection onto $\mathrm{Col}(B)$ is

$$P_W = B(B^T B)^{-1} B^T$$

The best data (closest point on W to $\mathbf{b}$) is:

$$P_W \mathbf{b} = \underbrace{B}_{\text{columns of } W} \underbrace{(B^T B)^{-1} B^T \mathbf{b}}_{\text{coefficients}}$$

Write $\hat{\mathbf{x}} = (B^T B)^{-1} B^T \mathbf{b}$. Then $P_W \mathbf{b} = B\hat{\mathbf{x}}$.

$\hat{\mathbf{x}}$ tells us the regression coefficients directly!

Our Ingredients

$$\underbrace{B}_{\text{perfect data matrix}} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \qquad \underbrace{\mathbf{b}}_{\text{observed data}} = \begin{pmatrix} 30 \\ 22 \\ 8 \end{pmatrix}$$

	col 1	col 2	data
$t = 0$	1	0	30
$t = 1$	1	1	22
$t = 2$	1	2	8

Column 1 = level marbles $(1, 1, 1)^T$. Column 2 = chopstick positions $(0, 1, 2)^T$.

What Does $\hat{\mathbf{x}}$ Mean?

Write $\hat{\mathbf{x}} = \begin{pmatrix} a \\ b \end{pmatrix}$. Then:

$$B\hat{\mathbf{x}} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ a+b \\ a+2b \end{pmatrix} = \begin{pmatrix} h(0) \\ h(1) \\ h(2) \end{pmatrix}$$

for the line $h(t) = a + bt$.

Finding $\hat{\mathbf{x}}$ gives us the intercept a and the slope b directly.

We don't need the 3×3 projection matrix P_W at all!

How to Compute $\hat{\mathbf{x}}$

We need:

$$\hat{\mathbf{x}} = \underbrace{(B^T B)^{-1}}_{\text{invertible!}} \underbrace{B^T \mathbf{b}}_{\text{a vector}}$$

Callback (Lectures 1–2): Apply **simultaneous row operations** to both matrices until the left one becomes I. The right side becomes $(B^T B)^{-1} B^T \mathbf{b}$.

Step 1: Compute $B^T B$

$$B^T B = \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}}_{B^T} \underbrace{\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}}_{B}$$

$$= \begin{pmatrix} 1+1+1 & 0+1+2 \\ 0+1+2 & 0+1+4 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}$$

A 2×2 symmetric matrix. (B has full column rank, so $B^T B$ is **invertible**.)

Step 2: Compute $B^T\mathbf{b}$

$$B^T\mathbf{b} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 30 \\ 22 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} 30 + 22 + 8 \\ 0 + 22 + 16 \end{pmatrix} = \begin{pmatrix} 60 \\ 38 \end{pmatrix}$$

Entry 1: sum of all measurements. Entry 2: weighted sum (by chopstick positions).

Step 3: Simultaneous Row Operations

Setup:

$$\hat{\mathbf{x}} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 60 \\ 38 \end{pmatrix}$$

Step 1: $r_2 \to r_2 - r_1$

$$= \begin{pmatrix} 3 & 3 \\ 0 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 60 \\ -22 \end{pmatrix}$$

Step 2: $r_2 \to \frac{1}{2} r_2$

$$= \begin{pmatrix} 3 & 3 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 60 \\ -11 \end{pmatrix}$$

Step 3: Simultaneous Row Operations (continued)

Step 3: $r_1 \to r_1 - 3r_2$

$$\hat{\mathbf{x}} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 93 \\ -11 \end{pmatrix}$$

Step 4: $r_1 \to \frac{1}{3} r_1$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 31 \\ -11 \end{pmatrix} = \begin{pmatrix} \mathbf{31} \\ \mathbf{-11} \end{pmatrix}$$

$$\hat{\mathbf{x}} = \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 31 \\ -11 \end{pmatrix}$$

The Best Line

$$\hat{\mathbf{x}} = \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 31 \\ -11 \end{pmatrix}$$

The best-fitting line is:

$$\boxed{h(t) = 31 - 11t}$$

$$\begin{aligned} a &= 31 && \leftarrow \text{predicted height at } t = 0 \\ b &= -11 && \leftarrow \text{the candle loses 11 cm per hour} \end{aligned}$$

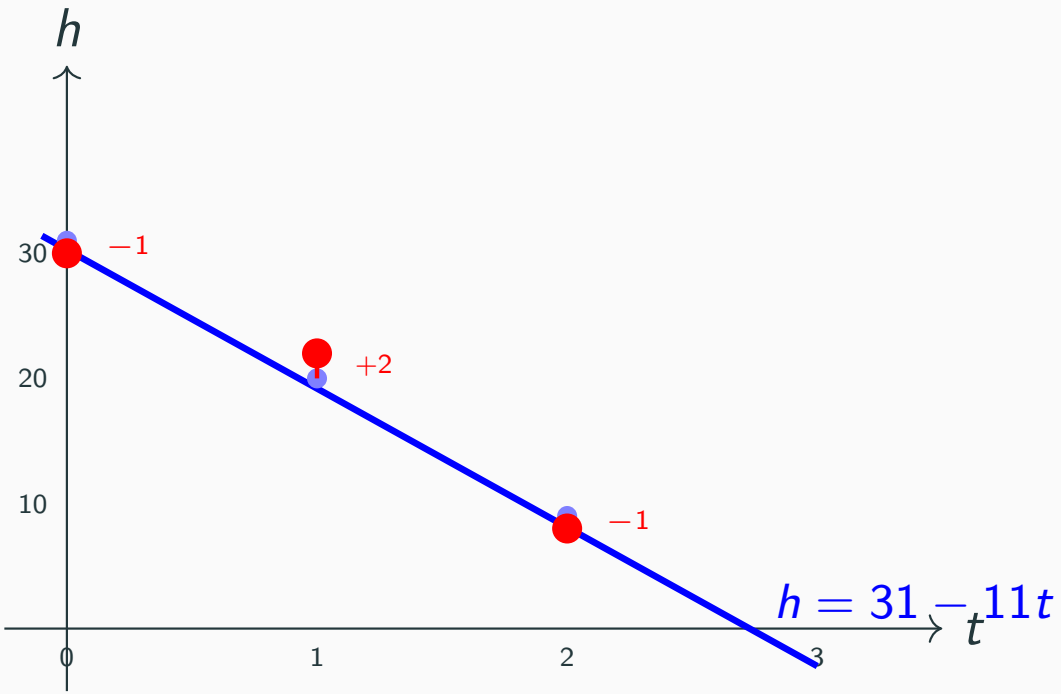
Check: Predicted vs. Actual

t	Actual h	Predicted $31 - 11t$	Error
0	30	31	-1
1	22	20	$+2$
2	8	9	-1

$$\text{Sum of squared errors} = (-1)^2 + 2^2 + (-1)^2 = 1 + 4 + 1 = \mathbf{6}$$

No line can do better than error $= 6$. This is the **minimum**.

The Best Line on the Scatter Plot



The line doesn't pass through any data point — but it's the **closest** any line can get.

Bonus: Find $(B^T B)^{-1}$

$$(B^T B)^{-1} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$r_2 \to r_2 - r_1, \quad r_2 \to \frac{1}{2} r_2, \quad r_1 \to r_1 - 3r_2, \quad r_1 \to \frac{1}{3} r_1$:

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} \frac{5}{6} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \frac{1}{6} \begin{pmatrix} \mathbf{5} & -3 \\ -3 & 3 \end{pmatrix}$$

Bonus: Cross-Fill $(B^T B)^{-1}$

Callback (Lecture 10, §4): Diagonally cross-fill $(B^T B)^{-1}$:

Pivot $(\mathbf{1}, \mathbf{1})$, value $= \frac{\mathbf{5}}{\mathbf{6}}$. Read off column 1: $\mathbf{d}_1 \propto \begin{pmatrix} 5 \\ -3 \end{pmatrix}$

$$B\mathbf{d}_1 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ -3 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$$

Remainder pivot at $(2, 2)$: $\mathbf{d}_2 \propto \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$B\mathbf{d}_2 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

Orthogonal Basis for the Perfect-Data Plane

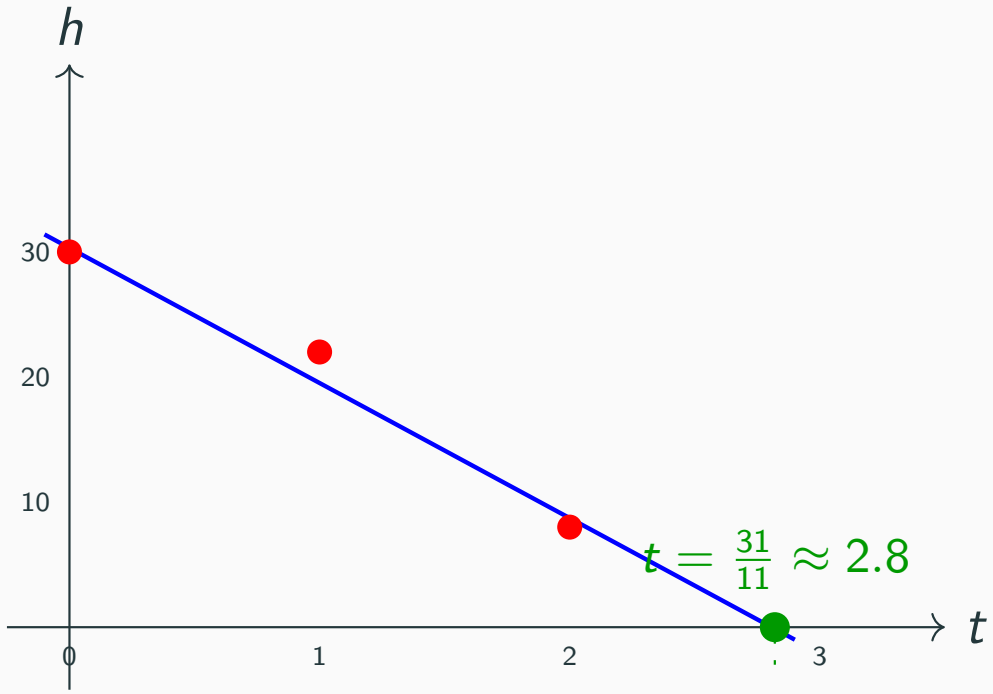
$$\begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}^T \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = 0 + 2 - 2 = 0 \quad \checkmark$$

$\left\{ \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}$ is an **orthogonal basis** for the perfect-data plane W.

(Same as Lecture 10, §4 example — same matrix B!)

Prediction: When Does the Candle Burn Out?

Our best line: $h(t) = 31 - 11t$.



Set $h(t) = 0$: $\quad 31 - 11t = 0 \quad \Longrightarrow \quad t = \frac{31}{11} \approx 2.8$ hours.

The candle burns out in about 2 hours 48 minutes.